Extending Affine Control Policies to Robust Control of Hybrid Systems (DC-DC Buck Converter Example)

Robin Vujanic, Marius Schmitt, Joe Warrington
Manfred Morari

Institut für Automatik (IfA)
Department of Electrical Engineering
Swiss Federal Institute of Technology (ETHZ)

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Outline

Introduction

Robust MPC for Linear Systems

Extension to Hybrid Systems

Example: DC-DC Buck Converter
- Linear Model
- Hybrid Model

Conclusions

Introduction

In this talk:

1. (Review of) MPC-based robust controllers for **linear systems**
2. **New robust controller** for some **hybrid models**
3. **Example**: DC-DC buck converter

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Linear Robust MPC

Finite horizon, optimal control problem formulation

$$
\begin{aligned}
\min \quad & \left[\sum_{k=0}^{N-1} (x_k - x_{ref})^T P (x_k - x_{ref}) + u_k^T Q u_k\right] \\
\text{s.t.} \quad & x_0 = \overline{x}_0, \\
& x_{k+1} = Ax_k + Bu_k \\
& x_k \in \mathcal{X}_k, \\
& u_k \in \mathcal{U}_k
\end{aligned}
$$

- ideally, obtain **optimal control policy** using e.g. DP
  - **"optimal decision is taken at each stage in the horizon"**
  - $u_k = \pi(x_k)$
  - often intractable
- conservative approximation: **open-loop control policy**
  - **"decide now for the entire horizon"** (plan can't be modifed)
  - $u_k = v_k$
  - poor performance, infeasibility issues
- middle ground: **affine recourse**
  - **"decisions are affinely adjusted once disturb. are meas."**
  -

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Finite horizon, robust optimal control problem formulation

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& \left.\begin{aligned} & x_{k+1} = Ax_k + Bu_k + Gw_k, \\ & x_k \in \mathcal{X}_k, \\ & u_k \in \mathcal{U}_k \end{aligned}\right\} \forall \mathbf{w} \in \mathcal{W}
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$$\begin{aligned} \min \quad & \mathbb{E}\left[(\mathbf{x}-\mathbf{x_{ref}})^T\mathbf{P}(\mathbf{x}-\mathbf{x_{ref}})+\mathbf{u}^T\mathbf{Q}\mathbf{u}\right] \\ \text{s.t.} \quad & \left.\begin{aligned} &\mathbf{x}=\mathbf{A}x_0+\mathbf{B}\mathbf{u}+\mathbf{G}\mathbf{w}, \\ &\mathbf{E_x}\mathbf{x}+\mathbf{E_u}\mathbf{u}\leq\mathbf{e} \end{aligned}\right\}\forall\mathbf{w}\in\mathcal{W} \end{aligned}$$

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  - $u_k=v_k+\sum_{j=0}^{k}K_{kj}x_j$

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Finite horizon, optimal control problem formulation

$$\begin{aligned} \min \quad & \mathbb{E}\left[(\mathbf{x}-\mathbf{x_{ref}})^T\mathbf{P}(\mathbf{x}-\mathbf{x_{ref}})+\mathbf{u}^T\mathbf{Q}\mathbf{u}\right] \\ \text{s.t.} \quad & \left.\begin{aligned} &\mathbf{x}=\mathbf{A}x_0+\mathbf{B}\mathbf{u}+\mathbf{G}\mathbf{w}, \\ &\mathbf{E_x}\mathbf{x}+\mathbf{E_u}\mathbf{u}\leq\mathbf{e} \end{aligned}\right\}\forall\mathbf{w}\in\mathcal{W} \end{aligned}$$

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  - **"decisions are affinely adjusted once disturb. are meas."**
  - $u_k = v_k + \sum_{j=0}^{k} M_{kj} w_j$

Explicit Robust Counterpart

- can show that the robust counterpart is equivalent to

$$\boxed{\begin{array}{ll} \min_{\mathbf{v},\mathbf{M},\Lambda} & \mathbb{E}\left[(\mathbf{x}-\mathbf{x_{ref}})^T\mathbf{P}(\mathbf{x}-\mathbf{x_{ref}})+\mathbf{u}^T\mathbf{Q}\mathbf{u}\right] \\ s.t. & \mathbf{E_x}\mathbf{A}x_0+(\mathbf{E_x}\mathbf{B}+\mathbf{E_u})\mathbf{v}-\mathbf{e}\leq -\Lambda^T\cdot h \\ & \Lambda^T\cdot S=\mathbf{E_x}\mathbf{G}+\mathbf{E_x}\mathbf{B}\mathbf{M}+\mathbf{E_u}\mathbf{M} \\ & \Lambda\geq 0 \end{array}} \qquad \text{(R-MPC}_{avg}\text{)}$$

  where w_k is bounded by $S\cdot w_k\leq h$ (Goulart 2006)
- finite convex QP

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Robust Control of Hybrid Systems

- hybrid system dynamics (MLD)

$$
\begin{aligned}
&x_{k+1} = Ax_k + Bu_k + B_2\delta_k + B_3 z_k + Gw_k \\
&E_x x_k + E_u u_k + E_\delta \delta_k + E_z z_k \leq e_k \\
&\delta_k \in \{0,1\}^{n_\delta}, z_k \in \mathbb{R}^{n_z}
\end{aligned}
$$

- δ_k, z_k characterize hybrid behavior,
  - in the dynamics, e.g. switching between *modes*
  - in the constr., e.g. logic conditions on the inputs
- wish to obtain a solution (with some recourse) to

$$
\begin{aligned}
\min_{\mathbf{x},\mathbf{u},\delta,\mathbf{z}} \quad & \mathbb{E}\left[(\mathbf{x}-\mathbf{x_{ref}})^T\mathbf{P}(\mathbf{x}-\mathbf{x_{ref}}) + \mathbf{u}^T\mathbf{Q}\mathbf{u}\right] \\
\text{s.t.} \quad & \mathbf{x} = \mathbf{A}x_0 + \mathbf{B}\mathbf{u} + \mathbf{B_2}\boldsymbol{\delta} + \mathbf{B_3}\mathbf{z} + \mathbf{G}\mathbf{w}, \\
& \mathbf{E_x}\mathbf{x} + \mathbf{E_u}\mathbf{u} + \mathbf{E}_{\boldsymbol{\delta}}\boldsymbol{\delta} + \mathbf{E_z}\mathbf{z} \leq \mathbf{e}\,, \qquad \forall \mathbf{w} \in \mathcal{W} \\
& \boldsymbol{\delta} \in \{0,1\}^{N\cdot n_\delta}
\end{aligned}
\tag{RHOCP}
$$

Affine recourse on continuous inputs - R-MPC$_{hyb}$

Main Idea

Proposed idea:

- we need recourse, but affine functions cannot easily provide binary inputs
- split the inputs into continuous inputs u and binary inputs d

$$x_{k+1} = A \cdot x_k + B_{cont} u_k + B_{bin} d_k + B_2 \cdot \delta_k + B_3 z_k + G w_k$$

- introduce affine recourse on the continuous inputs

$$\mathbf{u} := \mathbf{M} \cdot \mathbf{w} + \mathbf{v}$$

Assumption on G:

- disturbances only affect the continuous dynamics

Affine recourse on continuous inputs - R-MPC$_{hyb}$

- can show that the robust counterpart is

$$\boxed{\begin{array}{rl} \min\limits_{\mathbf{v},\mathbf{M},\mathbf{d},\boldsymbol{\delta},\mathbf{z},\Lambda} & \hat{f} + \mathrm{trace}(\mathbf{D}\cdot\mathbf{C_w}) \\ \text{s.t.} & \hat{\mathbf{e}} \leq -\Lambda^T h\,, \\ & \Lambda^T\mathbf{S} = \mathbf{E_x}\mathbf{B_u}\mathbf{M} + \mathbf{E_x}\mathbf{G} + \mathbf{E_u}\mathbf{M} \\ & \Lambda \geq 0 \quad \text{element-wise} \\ & \boldsymbol{\delta} \in \{0,1\}^{N\cdot n_\delta},\ \mathbf{d} \in \{0,1\}^{N\cdot n_d} \end{array}} \qquad \text{(R-MPC}_{hyb}\text{)}$$

- with

$$\hat{\mathbf{x}} \doteq \mathbf{A}x_0 + \mathbf{B_u}\mathbf{v} + \mathbf{B_d}\mathbf{d} + \mathbf{B_2}\boldsymbol{\delta} + \mathbf{B_3}\mathbf{z}$$
$$\hat{\mathbf{e}} \doteq \mathbf{E_x}\hat{\mathbf{x}} + \mathbf{E_u}\mathbf{v} + \mathbf{E_d}\mathbf{d} + \mathbf{E}_\delta\boldsymbol{\delta} + \mathbf{E_z}\mathbf{z} - \mathbf{e}$$

  (variables in case of zero disturbance)

- and appropriate $\mathbf{D}$

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Plant: the Buck Converter (BC) (1/2)

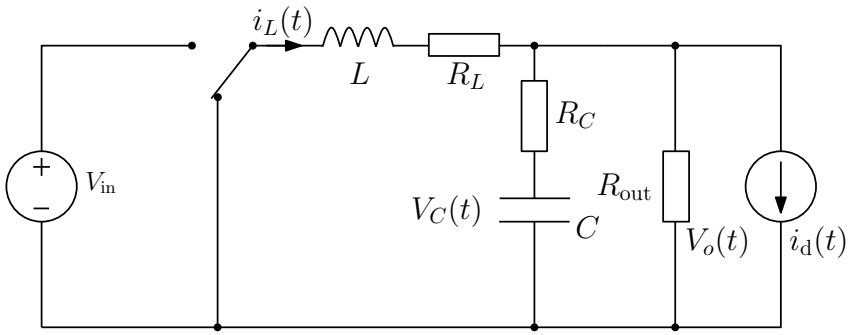
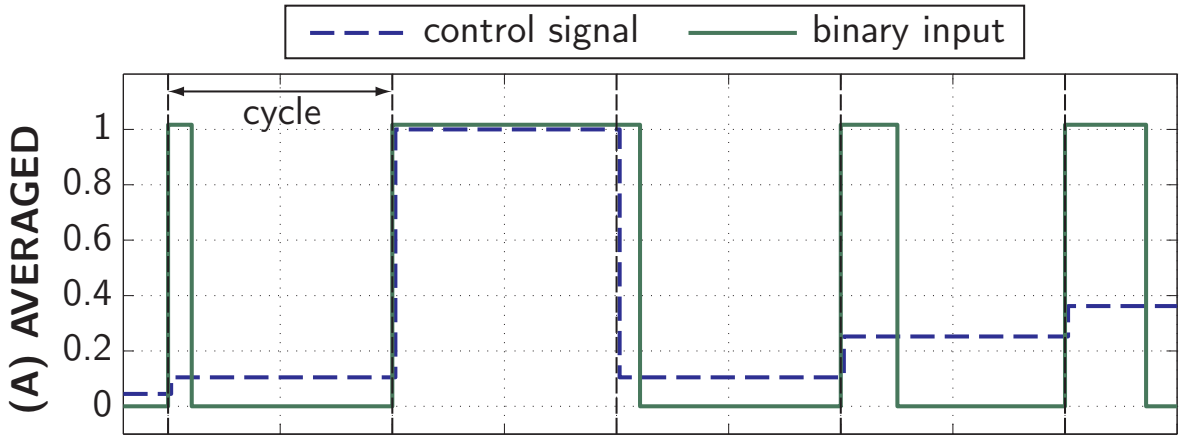


Figure: DC-DC buck converter circuit

- BC regulates input voltage V_{in} down to desired $V_{o,\text{ref}}$
- operated by switch (controlled input)
- disturbances: $|i_d| \leq 0.5 \cdot i_{L,\text{ref}}$
- state constraints: $i_L \leq 2 \cdot i_{L,\text{ref}}$
- sampling frequency: 10 kHz

Averaged Model (standard method)

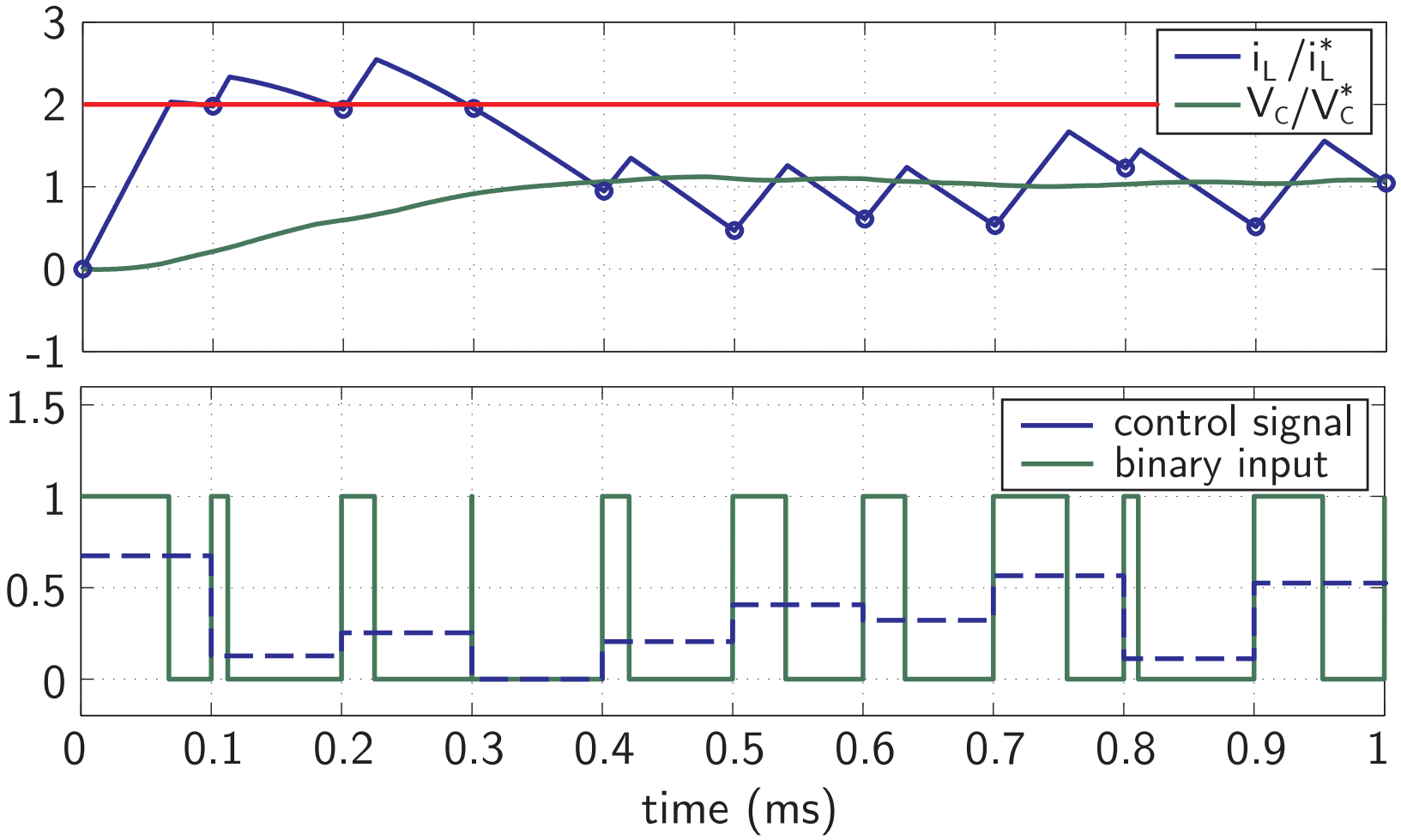


- replace $\delta(t) \in \{0, 1\}$ by duty cycle $d(t) \in [0, 1]$
- average dynamics "when off" and "when on" weighted by $d(t)$, obtain

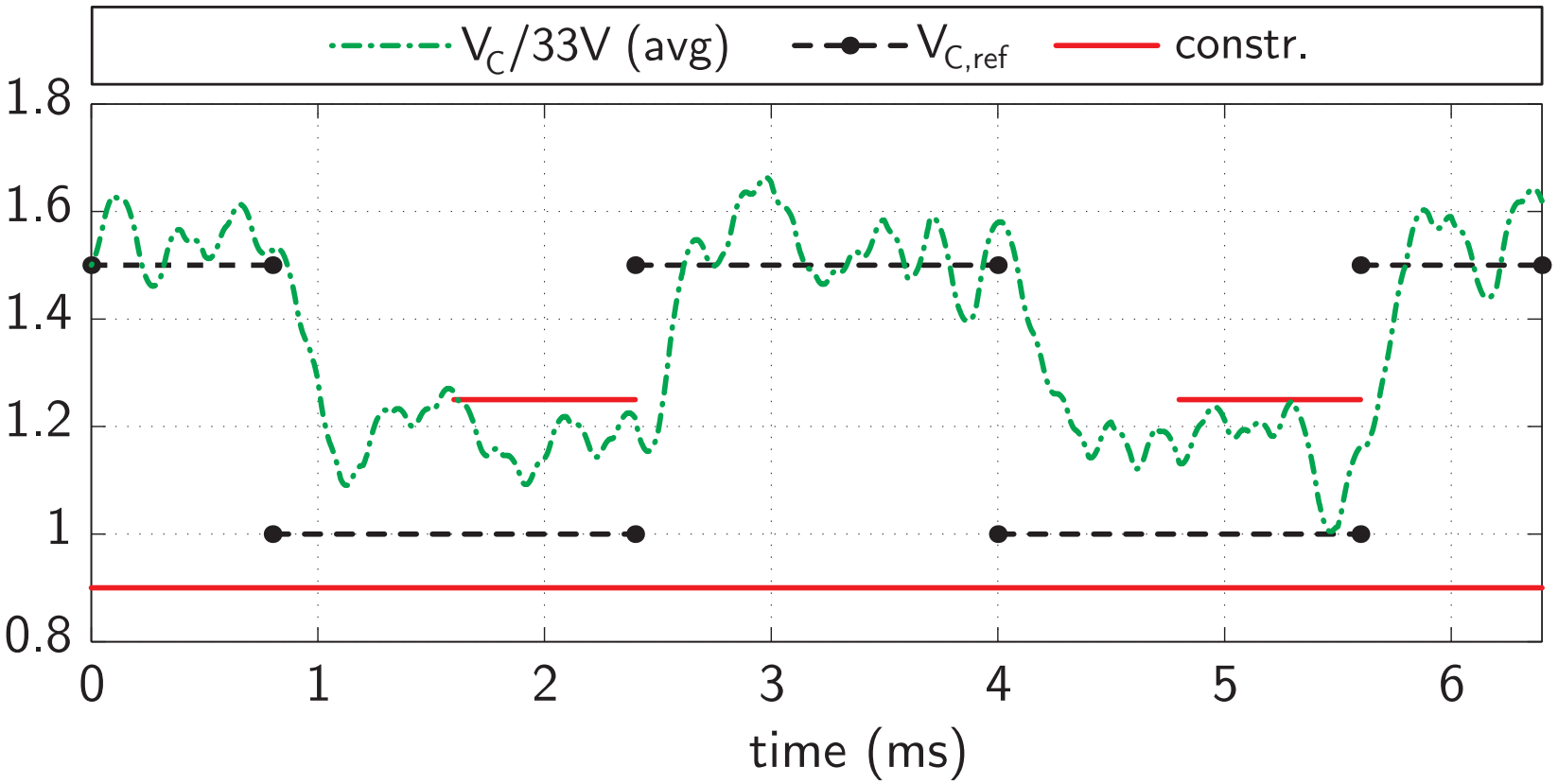
$$\dot{x} = Ax + Bu + Gw$$
$$u \in [0, 1]$$

- **linear model**

Performance of R-MPC$_{avg}$



R-MPC$_{avg}$ – Second Experiment



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Hybrid Model [2] (1/2)

- **Closer approximation of the switch**
- Divide time into cycles with M samples each
- New binary inputs: switch on $\delta_k^+ = 1$, switch off $\delta_k^- = 1$
- New continuous input: $u_{c,k}$
- New auxiliary state: s_k (integration of $\delta^+ - \delta^-$)
- switch position given by $s_k + u_{c,k}$

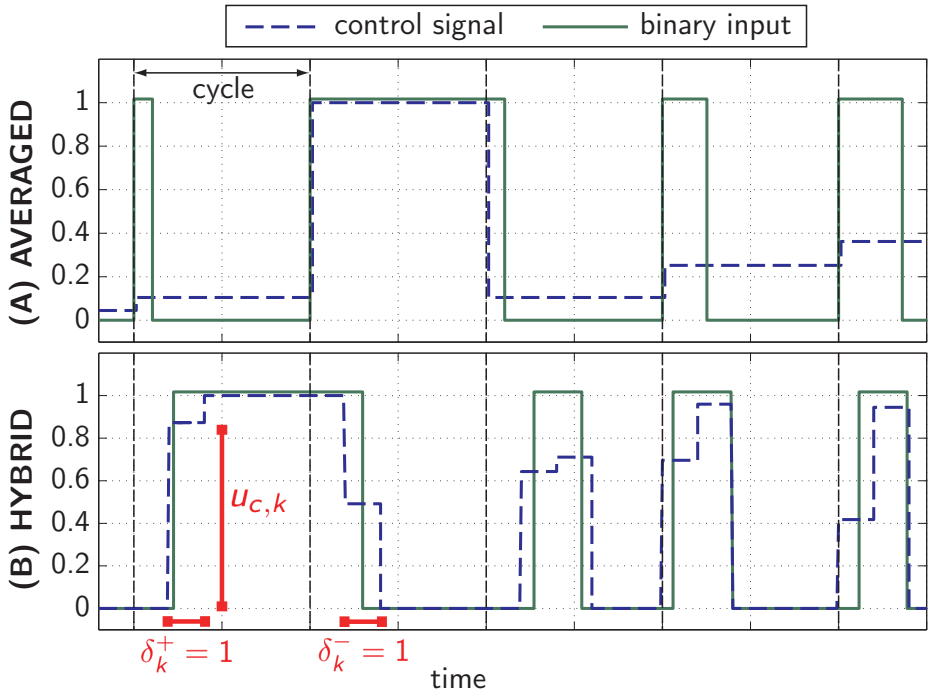


Figure: PWM in the hybrid model

Hybrid Model [2] (2/2)

- New system equations:

$$\begin{pmatrix} i_L \\ V_C \\ s \end{pmatrix}_{k+1} = \begin{pmatrix} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} i_L \\ V_C \\ s \end{pmatrix}_k + \begin{pmatrix} B_1 & 0 & 0 \\ B_2 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} u_c \\ \delta_+ \\ \delta_- \end{pmatrix}_k + \begin{pmatrix} G_1 \\ G_2 \\ 0 \end{pmatrix} w_k$$

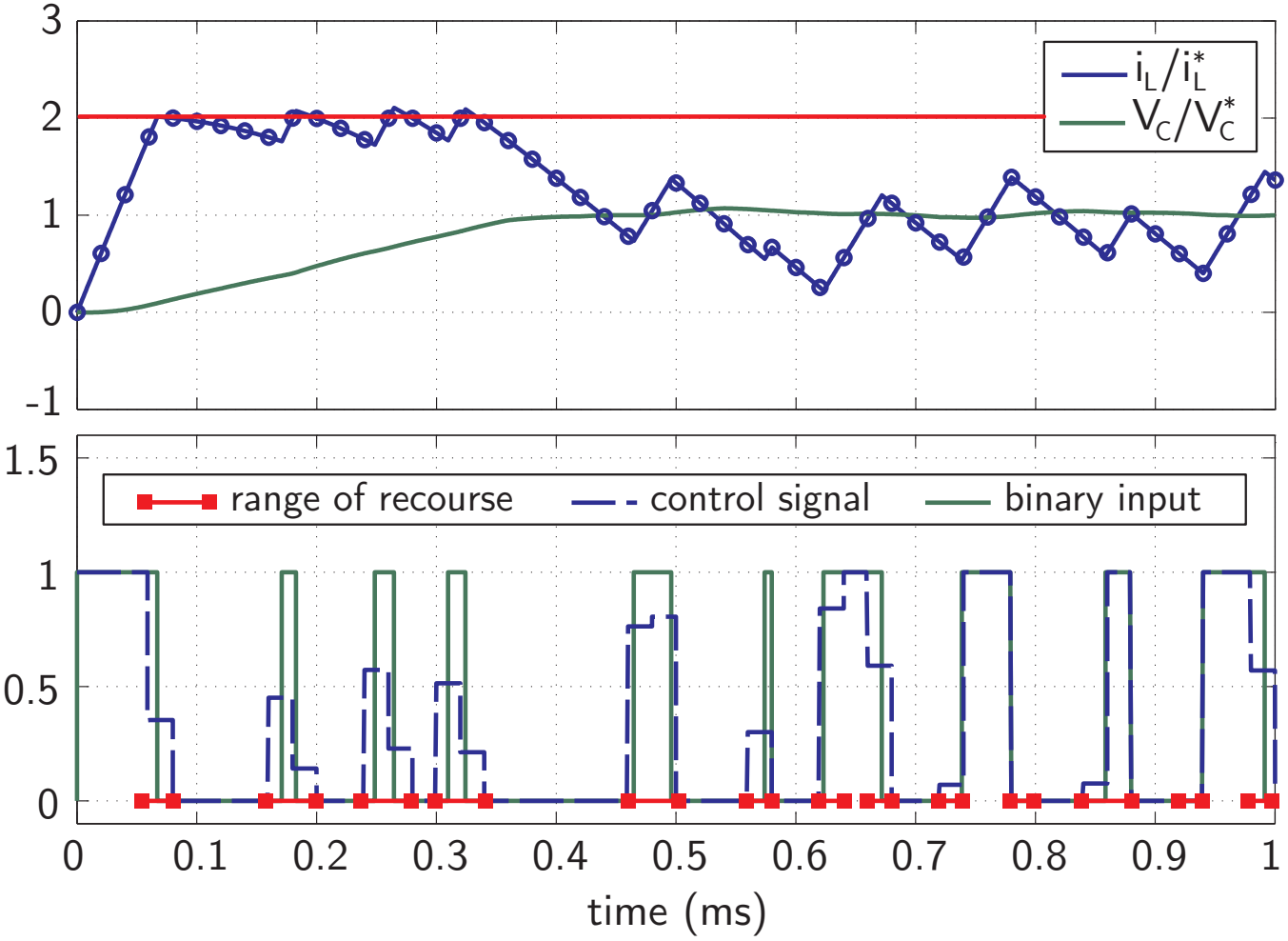
- New constraints:

$\delta_k^+, \delta_k^- \in \{0, 1\}$	binary inputs
$0 \leq s_k \leq 1$	binary state
$0 \leq s_k + u_{c,k} \leq 1$	limited input
$-\delta_k^- \leq u_{c,k} \leq \delta_k^+$	switching time
$\left.\begin{array}{l} \sum_{i=0}^{M-1} \delta_{k+i}^+ \leq 1 \\ \sum_{i=0}^{M-1} \delta_{k+i}^- \leq 1 \end{array}\right\}$	switching constraints

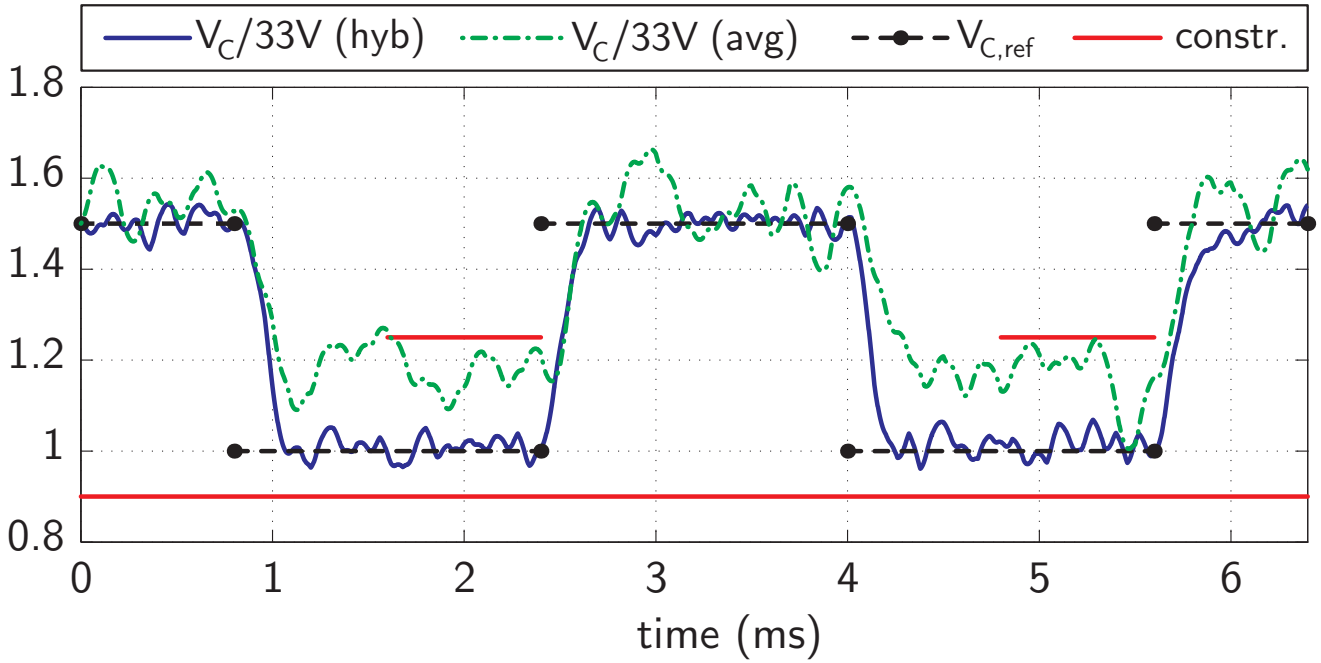
- together with $i_{L,k} \leq 2 \cdot i_{L,\text{ref}}$ they form

$$x_{k+1} = A \cdot x_k + B_{cont} u_k + B_{bin} d_k + B_2 \cdot \delta_k + B_3 z_k + G w_k$$
$$E_x x_k + E_u u_k + E_d d_k + E_\delta \delta_k + E_z z_k \leq e_k$$

Performance & Simulation results – R-MPC$_{hyb}$



R-MPC$_{hyb}$ – Second Experiment



Controller	RMS deviation of $V_o(t)$ 1st experiment	RMS deviation of $V_o(t)$ 2nd experiment
MPC$_{avg}$	2.09V	15.2$[V]$
MPC$_{hyb}$	0.88V	9.74$[V]$
MPC$_{hyb}^{open\ loop}$	0.89V	Infeasibility encountered

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- New robust controller R-MPC$_{hyb}$ for hybrid systems
- Assessment on the BC
  - good performance
  - no infeasibility issues
- Future work:
  - faster computation of solutions,
    e.g. using exclusively convex programming (DONE)
  - other applications?
    e.g. dynamic system at the output stage

Conclusions

- Questions?
- References:

1. Paul J. Goulart, Eric C. Kerrigan and Jan M. Maciejowski. Optimization over state feedback policies for robust control with constraints. 2006, Automatica 42 (523 - 533)
2. Claudia Fischer, Sebastian Mariethoz and Manfred Morari. Multisampled Hybrid Model Predictive Control for Pulse-Width Modulated Systems. 50th Conference on Decision and Control and European Control Conference (CDC-ECC), Dec 2011, Orlando, FL, USA
3. Paul J. Goulart and Eric C. Kerrigan. Input-to-state stability of robust receding horizon control with an expected value cost. 2008, Autimatica 22 (1171 - 1174)
4. Aharon Ben-Tal, Laurent El Ghaoui and Arkadi Nemirovski. Robust Optimization. 2009, Princeton Series in Applied Mathematics

- Slides of the talk on http://control.ee.ethz.ch/∼ vujanicr/
  - under "publications → talks"

Derivation of $\text{MPC}_{ol/cl}$

Robust constraints satisfaction

$$
\begin{aligned}
& \mathbf{E_x x} + \mathbf{E_u u} + \mathbf{E_d d} + \mathbf{E}_\delta \delta \leq \mathbf{e} && \forall \mathbf{w} \in \mathcal{W} \\
\Leftrightarrow \quad & \hat{\mathbf{e}} + (\mathbf{E_x B_u M} + \mathbf{E_x G} + \mathbf{E_u M})\mathbf{w} \leq 0, && \forall \mathbf{w} \in \mathcal{W} \\
\Leftrightarrow \quad & \hat{\mathbf{e}} + \max_{\mathbf{w} \in \mathcal{W}} (\mathbf{E_x B_u M} + \mathbf{E_x G} + \mathbf{E_u M})\mathbf{w} \leq 0
\end{aligned}
$$

with

$$
\begin{aligned}
\hat{\mathbf{x}} &\doteq \mathbf{A} x_0 + \mathbf{B_u v} + \mathbf{B_d d} + \mathbf{B_2} \boldsymbol{\delta} + \mathbf{B_3 z} \\
\hat{\mathbf{e}} &\doteq \mathbf{E_x} \hat{\mathbf{x}} + \mathbf{E_u v} + \mathbf{E_d d} + \mathbf{E}_{\boldsymbol{\delta}} \boldsymbol{\delta} + \mathbf{E_z z} - \mathbf{e}.
\end{aligned}
$$

Then

$$
\begin{aligned}
& \max_{\mathbf{w} \in \mathcal{W}} \quad (\mathbf{E_x B_u M} + \mathbf{E_x G} + \mathbf{E_u M})\mathbf{w} \ \text{ s.t. } \ \mathbf{S w} \leq \mathbf{h} \\
= & \min_{\Lambda \geq 0} \quad \Lambda^T \mathbf{h} \ \text{ s.t. } \ \Lambda^T \mathbf{S} = \mathbf{E_x B_u M} + \mathbf{E_x G} + \mathbf{E_u M}
\end{aligned}
$$

hence robust state constraint satisfaction equivalent to

$$
\begin{aligned}
& \hat{\mathbf{e}} \leq -\Lambda^T h\,, \\
& \Lambda^T \mathbf{S} = \mathbf{E_x B_u M} + \mathbf{E_x G} + \mathbf{E_u M}\,, \\
& \Lambda \geq 0 \quad \text{element-wise.}
\end{aligned}
$$

Derivation of Objective (Hybrid, OL case)

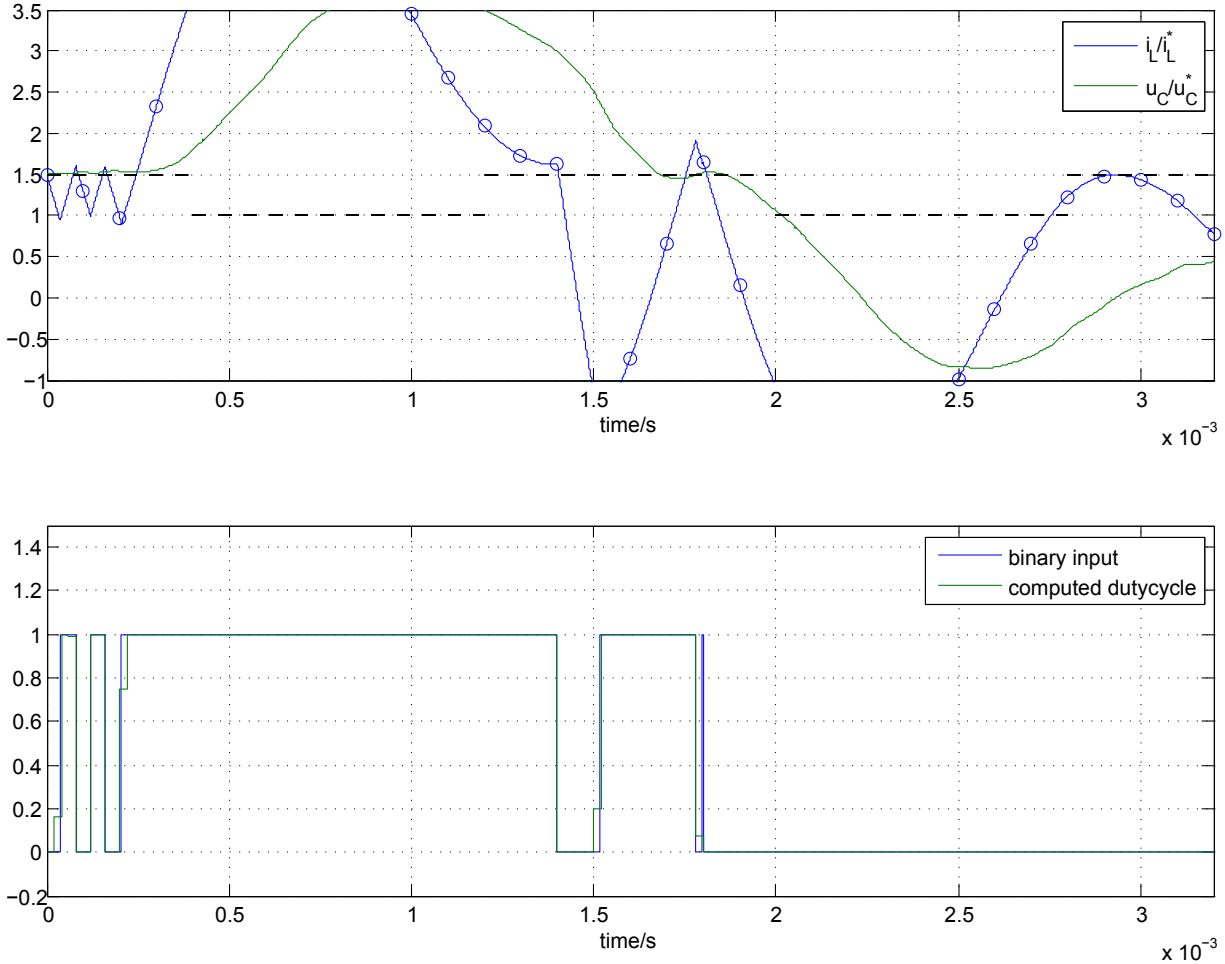
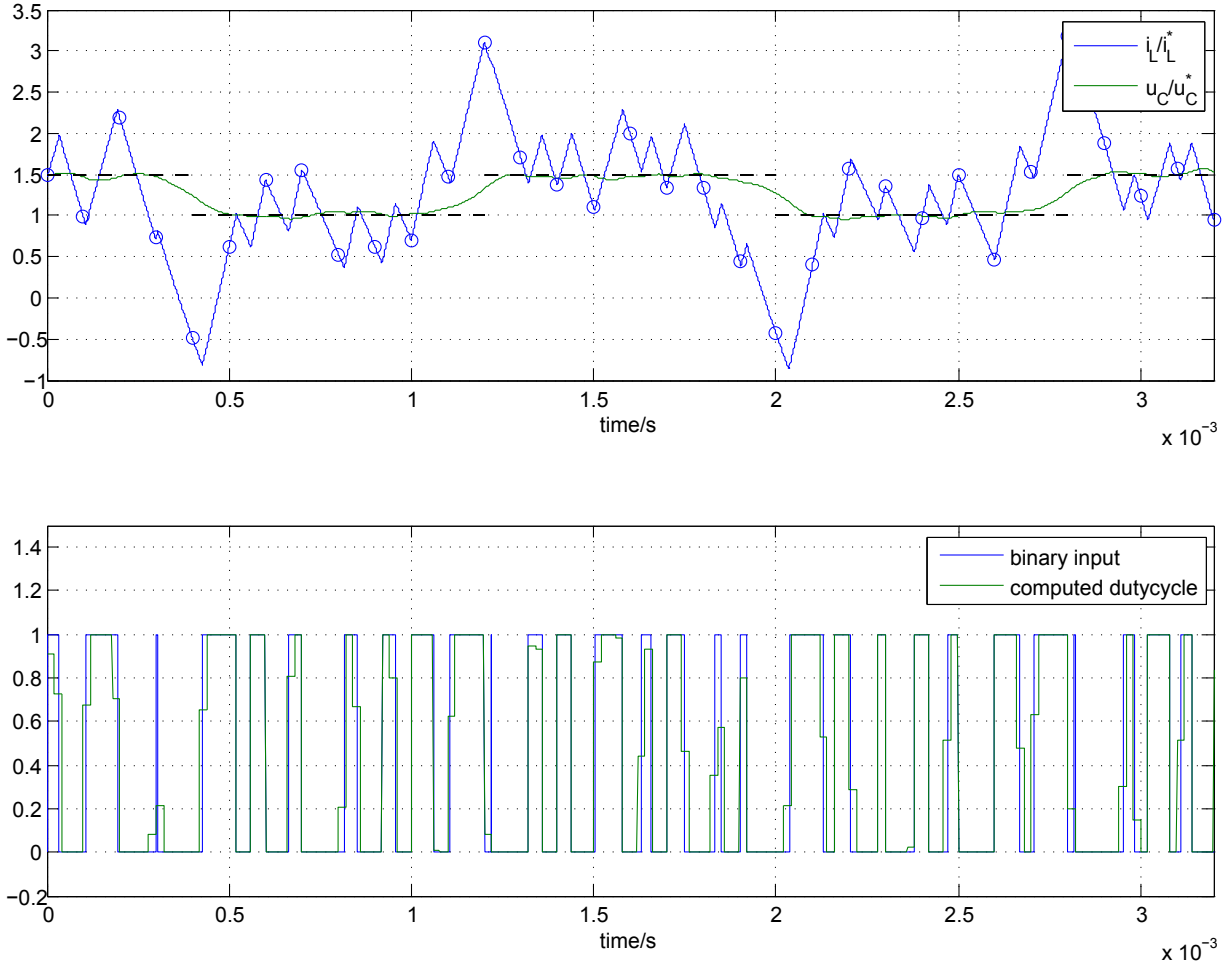
The output voltage is a linear function of the state x_k and the disturbances w_k

$$V_o = \frac{R_C R_{\text{out}}}{R_C + R_{\text{out}}} i_L + \frac{R_{\text{out}}}{R_C + R_{\text{out}}} V_C + \frac{R_C R_{\text{out}}}{R_C + R_{\text{out}}} i_d \qquad (1)$$

Assuming the disturbance is zero mean and iid.

$$\begin{aligned}
&\min_{\mathbf{x},\mathbf{u},\delta} f(\mathbf{x},\mathbf{u}) = \\
&= \min_{\mathbf{u},\delta} E[(\mathbf{A}x_0 + \mathbf{Bu} + \mathbf{B}_2\delta + \mathbf{B_3z} + \mathbf{Gw} - \mathbf{x_{ref}})^T\mathbf{P}\ldots \\
&(\mathbf{A}x_0 + \mathbf{Bu} + \mathbf{B}_2\delta + \mathbf{B_3z} + \mathbf{Gw} - \mathbf{x_{ref}}) + \mathbf{u}^T\mathbf{Qu}] \\
&= \min_{\mathbf{u},\delta}(\mathbf{A}x_0 + \mathbf{Bu} + \mathbf{B_2}\delta - \mathbf{x_{ref}})^T\mathbf{P}\ldots \\
&(\mathbf{A}x_0 + \mathbf{Bu} + \mathbf{B_2}\delta - \mathbf{x_{ref}}) + \mathbf{u}^T\mathbf{Qu}\ldots \\
&+\underbrace{2 \cdot E[(\mathbf{A}x_0 + \mathbf{Bu} + \mathbf{B_2}\delta - \mathbf{x_{ref}})^T\mathbf{PGw}]}_{=0} + E[\mathbf{w}^T\mathbf{G}^T\mathbf{PGw}] \\
&= \min_{\mathbf{u},\delta} f_{nom}(\mathbf{x},\mathbf{u}) + \text{const.}
\end{aligned}$$

Alternative approach # 1: ignoring disturbances



Alternative approach # 2: soft constraints